3 Points Each

2001

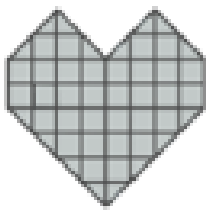
1. Julia has four candles. Each of her candles burns itself out in three hours. Julia lit two candles and placed them next to an open window. 30 minutes later the wind blew out one candle, and an hour later the wind blew out the second candle. Then Julia closed the window and lit all four candles. How long after this moment will the last candle burn out?

(A) 1 hr 30 min
(B) 2 hr
(C) 3 hr
(D) 7 hr 30 min
(E) 8 hr

2. Joseph had 7 sticks. He broke one of them into two pieces. How many sticks does Joseph have now?

(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

3. Kuba bought a chocolate heart for his mother (see the picture). Each chocolate square weighs 10 grams. What is the weight of the whole heart?



(A) 340 grams
(B) 360 grams
(C) 380 grams
(D) 400 grams
(E) 420 grams

4. There are 12 pairs of shoes on each of 10 shelves in the animal shoe store. 5 centipedes are the first customers. Three of them buy 30 pairs of shoes each, and two of them buy 5 pairs of shoes each. How many pairs of shoes are left on the shelves?

(A) 10
(B) 15
(C) 20
(D) 25
(E) 30

5. For five days, Kaya was helping her mother pick berries. On the first day, she ate most of her berries and gave her mother only one cup of berries. She decided that each day she would give her mother twice as much berries as the day before. How many cups of berries did Kaya give her mother over five days?

(A) 5
(B) 31
(C) 21
(D) 11
(E) 16

6. Which of the following expressions is correct?

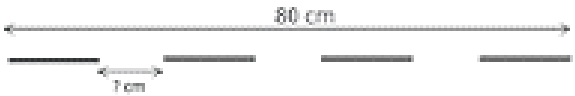
(A) $12+(4+8)=11$
(B) $8\times2+3=40$
(C) $2\times3+4x5=50$
(D) $(10+8)+2=14$
(E) $18-6+3=16$

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7. There are 19 girls and 12 boys in the school yard. At least how many students need to join them so that six groups with the same number of students can be formed?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

8. Four sticks, each 14 cm long, were placed in the way shown in the picture, for a total length of 80 cm. The distances between the sticks are equal. How long is each of these distances?



(A) 1 cm
(B) 2 cm
(C) 3 cm
(D) 5 cm
(E) 8 cm

4 Points Each

9. Bobby was born on Abby's third birthday. How many years later will Abby be twice as old as Bobby?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

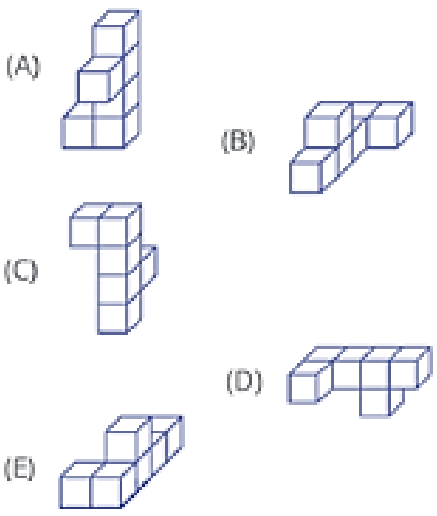
10. The lowest point of the Sniezna Cave in the Tatra Mountains is located 221 m below the cave entrance, and the highest point of the Sniezna Cave is located 419 m above the cave entrance. What is the depth (the distance between the lowest and highest points) of this cave?

(A) 198 m
(B) 221 m
(C) 419 m
(D) 640 m
(E) 650 m

11. I divided 20 pieces of candy among several children. Each child received at least one piece of candy, and everyone received a different number of pieces candy. What is the greatest possible number of children who received candy?

(A) 20
(B) 10
(C) 8
(D) 6
(E) 5

12. Which figure is different from all the others?



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13. Ella and Bella got on a super-train. Ella took a seat in the seventeenth car counting from the front of the train, and Bella was seated in the thirty-fourth car counting from the end. The girls were sitting in the same car. How many cars did the super-train have?

(A) 48
(B) 49
(C) 50
(D) 51
(E) 52

14. Adam and Bart are picking up chestnuts. At a certain moment they have the same number of chestnuts, and then Adam gives Bart half of all his chestnuts. How many times is the number of chestnuts Bart has greater than the number of chestnuts Adam has now?

(A) 2 times
(B) 3 times
(C) 4 times
(D) 5 times
(E) It depends on the number of chestnuts they had at the beginning.

15. There are squares and triangles on the table. They have 17 vertices altogether. How many triangles are there on the table?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

16. What is the least number of matches that must be added to the picture in order to get exactly 11 squares?

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6



5 Points Each

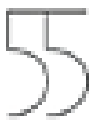
17. In a certain picture you can see numbers

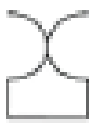


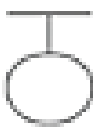
with their mirror reflections.



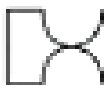
What is the next picture in the sequence of reflections?

(A) 

(B) 

(C) 

(D) 

(E) 

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18. Which of the five napkins below comes from this paper cut-out?



(A) 

(B) 

(C) 

(D) 

(E) 

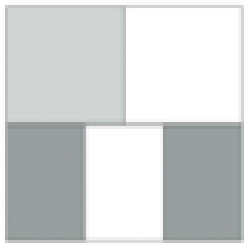
19. 12 boys and 8 girls were members of the Math Club. Every week new members were admitted: 1 boy and 2 girls. How many members will this club have when the number of boys equals the number of girls?

(A) 20
(B) 24
(C) 28
(D) 32
(E) 36

20. How many three-digit numbers are there such that the sum of their digits equals 4?

(A) 10
(B) 9
(C) 8
(D) 7
(E) 6

21. Five girls made a square with their beach towels (see the picture), The towels of Anita and Beth are squares with a perimeter of 720 cm each. The towels of Cindy, Debbie, and Eva are identical rectangles. What is the perimeter of one of these rectangles?



(A) 600 cm
(B) 560 cm
(C) 440 cm
(D) 360 cm
(E) 300 cm

22. Todd has as much money as Will and Kevin have together. Will has 10 dollars less than Kevin. The three boys have 40 dollars altogether. How much money does Will have?

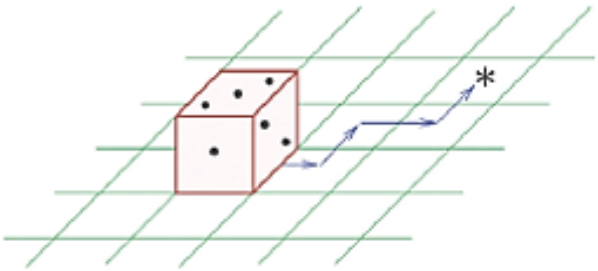
(A) $4
(B) $5
(C) $10
(D) $15
(E) $20

23. I have 11 candy bars in each of three baskets. I take out one candy bar from each basket in the following order: from the left, from the middle, from the right, from the middle, from the left, from the middle, from the right, and so on. What is the largest number of candy bars left in one of the baskets when the middle basket is empty?

(A) 1
(B) 2
(C) 5
(D) 6
(E) 11

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24. The sum of the dots on the opposite sides of a die is seven. We move a die on a grid as the picture shows. The die is rotated once for each square as it is moved as shown by the arrows. How many dots are on the top of the die when it is located on the square marked with*?



(A) 5
(B) 4
(C) 3
(D) 1
(E) other number

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Answers key:

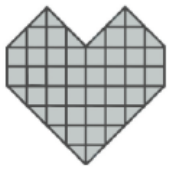
1. C
2. D
3. D
4. C
5. B
6. E
7. E
8. E
9. C
10. D
11. E
12. E
13. C
14. B
15. C
16. A
17. C
18. C
19. D
20. A
21. A
22. B
23. D
24. E

3 Point Solutions

1 **(C) 3 hr**
Each candle burns for 3 hours. Julia lit only 2 candles at first, so there were 2 candles that were not lit at all before the window was closed. Since these two candles were "brand new" when lit after the window was closed, it will take 3 hours for these candles to burn out.

2 **(D) 8**
Joseph keeps 6 original unbroken sticks and adds 2 new sticks by breaking the seventh stick into two pieces, so he has 8 sticks now.

3 **(D) 400 grams**



Count all the full squares, $(2 + 4 + 5 + 5) \times 2 = 32$ counted by columns. There are 16 half-squares which is equal to 8 full squares. The area of the heart is $32 + 8 = 40$ squares altogether. Thus, the weight of the whole heart is 40×10 grams $= 400$ grams.

4 **(C) 20**
Altogether there were $12 \times 10 = 120$ pairs of shoes in the store. The centipedes bought $3 \times 30 + 2 \times 5 = 90 + 10 = 100$ pairs. Thus, there were $120 - 100 = 20$ pairs of shoes left.

5 **(B) 31**
Kaya gave her mother one cup of berries on the first day. Each day Kaya doubles how many cups of berries she gives her mother, so Kaya gave her mother $1 + 2 + 4 + 8 + 16 = 31$ cups of berries over 5 days.

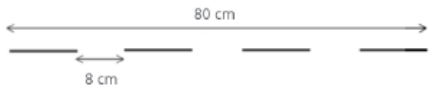
6 **(E) $18 - 6 \div 3 = 16$**
Remember to perform the operations in the correct order (multiplication and division first, from left to right, then addition and subtraction from left to right). Also remember that you have to perform the operations within parentheses first.

Calculations:

(A) $12 \div (4 + 8) = 12 \div 12 = 1 \neq 11$
(B) $8 \times 2 + 3 = 16 + 3 = 19 \neq 40$
(C) $2 \times 3 + 4 \times 5 = 6 + 20 = 26 \neq 50$
(D) $(10 + 8) \div 2 = 18 \div 2 = 9 \neq 14$
(E) $18 - 6 \div 3 = 18 - 2 = 16$

7 **(E) 5**
Together there are $19 + 12 = 31$ children. The smallest multiple of 6 greater than 31 is 36. $36 - 31 = 5$. So, 5 students need to join them.

8 **(E) 8 cm**
Altogether, the length of all 4 sticks is 4×14 cm $= 56$ cm. The total space between them is 80 cm – 56 cm = 24 cm. There are 3 gaps between the sticks, so each distance has to be 24 cm ÷ 3 = 8 cm.



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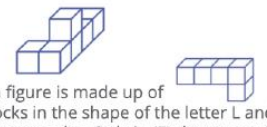
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4 Point Solutions

9 **(C) 3**
Abby will be twice Bobby's age when Bobby will be as old as the difference between their ages, which is 3. This will happen 3 years later, when Abby is 6 and Bobby is 3.

10 **(D) 640 m**
The depth of this cave is 221 m + 419 m = 640 m.

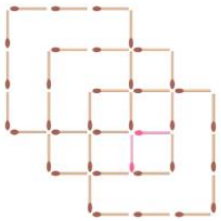
11 **(E) 5**
The 6 smallest and different numbers are 1, 2, 3, 4, 5, and 6. Their sum is 21 which exceeds 20, so you cannot distribute 20 pieces of candy according to these rules among 6 children. If you ignore the number 1, you can follow the rules and distribute 20 pieces of candy among 5 children; 2, 3, 4, 5, 6 are five different numbers and 2 + 3 + 4 + 5 + 6 = 20. So, the maximum number of children who received candy is 5.

12 **(E)**

Each figure is made up of 5 blocks in the shape of the letter L and one more cube. Only in (E) that one cube has a common edge with the corner cube of the piece shown.

13 **(C) 50**
By adding 17 and 34 we get 51 cars. The car in which both girls sat in was counted twice, so there were 51 – 1 = 50 cars.

14 **(B) 3 times**
Adam and Bart have the same number of chestnuts at the beginning and Adam can split his group of chestnuts into two halves, so Bart can do the same and together they have 4 halves of chestnuts. If Adam gives Bart one half of his chestnuts, then Adam will have 1 half and Bart will have 3 halves, so at the end Bart has 3 times as many chestnuts as Adam has.

15 **(C) 3**
Any square has 4 vertices and any triangle has 3 vertices. 17 is neither a multiple of 4 nor a multiple of 3, so there is at least one square and one triangle on the table. Together, one square and one triangle have 7 vertices, so there are still 10 vertices not assigned to any particular figure. 10 is neither a multiple of 4 nor a multiple of 3, so there is yet another pair of a square and a triangle on the table having another 7 vertices altogether. There are still 10 – 7 = 3 unassigned vertices, so the rest of vertices belong to a triangle. Hence, there are 3 triangles on the table.

16 **(A) 2**

There are 8 squares visible in the original picture: 3 are large (3 × 3), 2 are medium (2 × 2) and 3 are small (1 × 1). By adding the two matches shown here in red, 3 more small squares will be added for a total of 11 squares altogether.

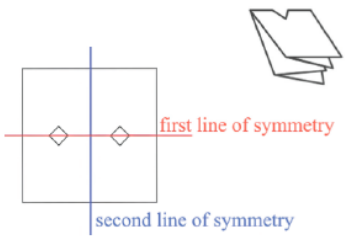
5 Point Solutions

17 **(C)**

The images in the problem have the digit first facing the opposite the direction. Answer (C) shows the number 5 in this way.

18 **(C)**

The top edge of the folded napkin is one line of symmetry and then, after the first unfolding, the edges on the left become one edge which is also a line of symmetry. In terms of the first line of symmetry, the paper cut-out looks like a diamond after the first unfolding and then, after unfolding along the second line of symmetry, we see two diamonds. After rotating the unfolded napkin by 90° we get the napkin shown in (C).



19 **(D) 32**
At the beginning, there were 12 – 8 = 4 more boys than girls. Since every week one more girl than boy was admitted, it will take 4 weeks for the numbers to be equal. After 4 weeks the number of girls will be 8 + 4 × 2 = 16 and the number of boys will be 12 + 4 × 1 = 16. The club will have 32 members then.

20 **(A) 10**
All the numbers that fulfill this requirement are (in the decreasing order): 400, 310, 301, 220, 211, 202, 130, 121, 112, and 103. There are 10 such numbers.

21 **(A) 600 cm**
The length of Anita's or Beth's towel is 720 cm ÷ 4 = 180 cm. Hence the side of the big square is 2 × 180 cm = 360 cm long. To get the width of one of the three smaller towels, divide 360 cm by 3, which is 120 cm. The length of one of these towels is 180 cm (half of the side of the big square). Thus, the perimeter of one rectangular towel is 2 × 120 cm + 2 × 180 cm = 600 cm.

22 **(B) $5**
Todd has 20 dollars and Will and Kevin together also have 20 dollars (20 + 20 = 40). Since Will has 10 dollars less than Kevin, he has 5 dollars and Kevin has 15 dollars (15 – 10 = 5; 5 + 15 = 20).

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23 (D) 6

The table below records the number of candy bars in each basket (L = left, M = middle, R = right) after consecutive actions until the middle basket is empty.

L	M	R	M	L	M	R	M	L	M	R
10	10	10	9	9	8	9	7	8	6	8

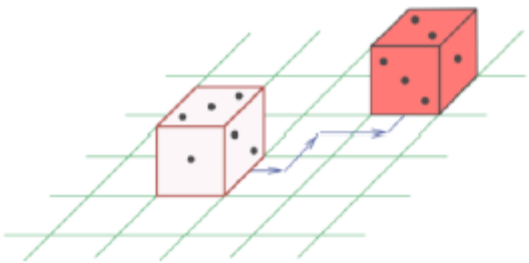
M	L	M	R	M	L	M	R	M	L	M
5	7	4	7	3	6	2	6	1	5	0

The right basket contains 6 candy bars and the left basket contains 5 candy bars at the moment when the middle basket is empty, so 6 is the largest number of candy bars left.

---

24 (E) other number

Remember that the sum of the numbers on the opposite sides of a die is 7. This tells us which numbers are on each of the sides that we cannot see in the picture. So,
after the first move we will have
1 in front, 5 on top, and 3 to the right;
after the second move we will have
2 in front, 1 on top, and 3 to the right;
after the third move we will have
2 in front, 4 on top, and 1 to the right;
after the fourth move we will have
3 in front, 2 on top, and 1 to the right.
The final position is 2 on top which is "other number" from the options listed.



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